

Exploring bounds of the 3 state misanthrope interacting particle system to ensure stationarity and translation-invariance, with other results

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Abstract

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1 Introduction

An **interacting particle system** (IPS) is a Markov chain, formed of a randomly interacting collection of nodes that occupy positions on a line, corresponding to those of $\mathbb{Z}$. These nodes each hold a number of particles, corresponding to elements of the node-wise state space $S \subseteq \mathbb{Z}$, and defined by some stochastic process. This means that an IPS has state space $S^{\mathbb{Z}}$. This paper will cover the IPS where particles in nodes jump to their nearest neighbours - the **nearest neighbour process** - but in particular, the case such that having more particles in a node means they jump to neighbours with greater frequency - the **misanthrope process**, introduced in Coccozza-Thivent [3].

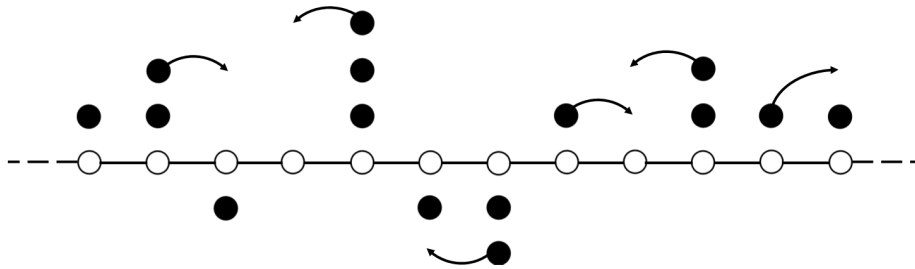


Figure 1: Representation of a Nearest Neighbor Interacting Particle System

As background information, know that much work has been done in the case of $|S| = 2$. Models with this property include the asymmetric simple exclusion process (ASEP), which ascribes constant left and right jump rates to the process, and the totally ASEP (TASEP), which sets one of the jump

rates to 0. Developed largely in the 80s, along with many others, in Liggett [7], these models have been the basis of research into topics such as travelling waves, as in Ferrari, Kipnis, and Saada [4] or tagged particles, as in Ferrari [5]. For more modern research see Huveneers and Simenhaus [6], in studying the asymptotic properties of a random walk. The advantages of the 2 state model is that particle jumps can be defined as a function of the indicator variable.

More complicated is the case of $|S| > 2$, where new variables must be introduced. We can use these higher orders to represent more complicated models, such for VoIP traffic modelling, as in Simionovici et. al [8] when it was used in prediction techniques to anticipate incoming web traffic.

In this case, there exist attractive particle systems, that try to avoid clustering on single nodes. There also exist IPSs that have a stationary product measure, which is a subset of those with a stationary distribution on the state space $S \subseteq \mathbb{Z}$ - these approach a finite mean value for each node as time goes on. Following on from the work in Balazs and Seppäläinen [1], we look to study those systems that are both attractive and have a stationary product measure. Among possible uses are constructing new models, such as the zero-range and bricklayer with exponentially increasing jump rates, as in Balázs, Rassoul-Agha, Seppäläinen, and Sethuraman [2].

In this paper, we demonstrate a general framework, with an example, that creates sufficient conditions for a 3 state misanthrope process to have a stationary product measure, without assuming particles jump left and right at the same rate.

2 The Model

A family of nearest neighbour IPSs are documented in section 1.2 of Balazs and Seppäläinen [1], which the reader is encouraged to read, since these give further clarity to some of the many ways an IPS may be constructed.

2.1 The IPS and the misanthrope Process

2.1.1 The misanthrope process

Paraphrasing from section 1.1 of Balazs and Seppäläinen. [1], the *misanthrope process* is a member of the nearest neighbour process family where, for $i \in \mathbb{Z}$, and for

$$-\infty \leq w^{\min} \leq 0 < w^{\max} \leq \infty \quad w^{\min}, w^{\max} \text{ in the extended integers} \quad (2.1)$$

we define the single site state space

$$S := \begin{cases} \{z \in \mathbb{Z} : w^{\min} \leq z \leq w^{\max}\} & \text{if } w^{\min} \text{ and } w^{\max} \text{ are finite} \\ \{z \in \mathbb{Z} : w^{\min} \leq z < w^{\max}\} & \text{if } w^{\min} \text{ is finite but } w^{\max} \text{ is not} \\ \{z \in \mathbb{Z} : w^{\min} < z \leq w^{\max}\} & \text{if } w^{\min} \text{ is not finite but } w^{\max} \text{ is} \\ \{z \in \mathbb{Z} : w^{\min} < z < w^{\max}\} & \text{if } w^{\min} \text{ and } w^{\max} \text{ are neither finite} \end{cases} \quad (2.2)$$

and the configuration space

$$\Omega = \{\underline{w} = (w_i)_{i \in \mathbb{Z}} : w_i \in S\} = S^{\mathbb{Z}} \quad (2.3)$$

to describe the positions and states available for particles in our IPS.

We let, for functions $q, p : S^2 \rightarrow [0, 1]$, a particle jump from a node i to its right neighbour, $i+1$, and a particle jump from a node $i+1$ to its left neighbour, i , with rates such that

$$\begin{aligned} (w_i, w_{i+1}) &\longrightarrow (w_i - 1, w_{i+1} + 1) \quad \text{with rate} \quad p(w_i, w_{i+1}) \\ (w_i, w_{i+1}) &\longrightarrow (w_i + 1, w_{i+1} - 1) \quad \text{with rate} \quad q(w_i, w_{i+1}) \end{aligned}$$

Conditionally on the continuous time Markov chain, $\underline{w}(t) \in \Omega$, these moves are independent of each other, and occur with independent exponential waiting times.

Further to this, we impose the following assumptions on the rates

Assumption 2.1 (Absorption at endpoints). *For $w^{\min}$ and $w^{\max}$ as before, when either is finite,*

$$p(w^{\min}, \cdot) \equiv q(\cdot, w^{\min}) \equiv p(\cdot, w^{\max}) \equiv q(w^{\max}, \cdot) \equiv 0 \quad (2.4)$$

where p and q are 0 everywhere else, or only one is equivalent to the 0 function (as in the totally asymmetric simple exclusion case).

If we imagine jumps into a node as a gain, and those out of it as a loss, with $w^{\min}$ as empty, and $w^{\max}$ as full, this can be viewed as saying that one cannot take from an empty bin, and one cannot overfill a bin.

Assumption 2.2 (Law of attractivity). *We have that*

$p(\cdot, \cdot)$ is **non-decreasing** in the 1st value, and **non-increasing** in the 2nd value

$q(\cdot, \cdot)$ is **non-increasing** in the 1st value, and **non-decreasing** in the 2nd value

For example, $p(5, 1) \geq p(2, 1)$, $q(1, 5) \geq q(1, 2)$

We can say that the nodes in the misanthrope process can only get so many or few particles, and tends to "smooth out" over time, as nodes do not wish to accumulate particles, so to speak. On a large scale, the misanthrope process behaves somewhat like temperature change over time in a wire for a closed system - the hot and cold positions eventually reach their average temperature.

Absorption at endpoints and the Law of attractivity define the whole family of misanthrope processes. We shall also assume the cyclic property and the symmetric property in our considerations, as stated in Coccozza-Thivent[3].

Assumption 2.3 (Cyclic property). *For all $x, y, z \in S$*

$$\begin{aligned} p(x, y) + p(y, z) + p(z, x) &+ q(x, y) + q(y, z) + q(z, x) = \\ = p(x, z) + p(z, y) + p(y, x) &+ q(x, z) + q(z, y) + q(y, x) \end{aligned}$$

which means that moving x to y to z is the same as moving from z to y to x .

Assumption 2.4 (Symmetric property). *There exist a function, $f : S \rightarrow \mathbb{R}$, and symmetric functions (ie, the order of input terms does not matter) $s_q : S^2 \rightarrow \mathbb{R}$ and $s_p : S^2 \rightarrow \mathbb{R}$, such that, for $w^{max} < \infty$*

$$\begin{aligned} 0 &= f(w^{max}) \\ q(y, z) &= f(z) \cdot s_q(y + 1, z) = f(z) \cdot s_q(z, y + 1) \\ p(y, z) &= f(y) \cdot s_p(y, z + 1) = f(y) \cdot s_p(z + 1, y) \end{aligned}$$

2.2 The infinitesimal generator

We shall now introduce the notion of the **infinitesimal generator**. Within the infinitesimal generator we hold all possible information about the the interacting particle system, and it is an important tool in describing the notion of a stationary product measure.

Definition 2.5 (Bounded cylinder function). *For, Ω as in 2.1.1, denote a particular state in Ω , by $\underline{\eta} = (\dots, \eta_{-1}, \eta_0, \eta_1, \dots)$. Then, $\varphi : \Omega \rightarrow \mathbb{R}$ is a **bounded cylinder function**, iff it depends only on a bounded finite set in $\underline{\eta}$, say $\{a, \dots, b\}$, leaving the complement alone.*

Definition 2.6 (Infinitesimal Generator). *For $\underline{w}, \underline{\eta}, \hat{\underline{\eta}}$ in Ω , we let $\underline{\eta}$ be some fixed state, $\underline{w} = (\dots, w_{-1}, w_0, w_1, \dots)$ describe some process, and $\hat{\underline{\eta}}$ be a placeholder for all possible states. Let $\varphi : \Omega \rightarrow \mathbb{R}$ be a bounded cylinder function.*

*Then an **infinitesimal generator**, $(L\varphi)$ is such that*

$$(L\varphi)(\underline{\eta}) = \sum_{\hat{\underline{\eta}}} \text{rate}(\underline{\eta} \rightarrow \hat{\underline{\eta}}) (\varphi(\hat{\underline{\eta}}) - \varphi(\underline{\eta})) \quad (2.5)$$

where $\text{rate}(\underline{\eta} \rightarrow \hat{\underline{\eta}})$ is some coefficient describing the particle jump rates that describe the movement from $\underline{\eta}$ to $\hat{\underline{\eta}}$.

It is worth noting that this definition is for all IPSs, not just those of nearest neighbour type.

The above definition works in general, but we must be more specific to apply it to our model. Let us now compute for nearest neighbour type particle jumps.

In Liggett [7], it is proven that if the rate of change of an infinitesimal generator is 0 on some bounded cylinder function that extends its mapping outside itself, then the whole measure is stationary. With the statements below, we.

Theorem 2.7. Let $\underline{w}, \underline{\eta}, \hat{\underline{\eta}}$ be as in Def 2.7, and $\varphi : \Omega \rightarrow \Omega$ be a particular test function, which is a bounded cylinder function on $\{a, \dots, b\}$. Denoting, also, $\underline{\eta}^{i,i+1}$ such that $(\underline{\eta}^{i,i+1})_j$ equals

$$\begin{cases} \eta_j & \text{if } j \neq i, i+1 \\ \eta_i - 1 & \text{if } j = i \\ \eta_i + 1 & \text{if } j = i+1 \end{cases}$$

Then

$$\frac{d}{dt} \mathbb{E}(\varphi(\underline{w}(t))) = \mathbb{E}((L\varphi)(\underline{w}(t))) \quad (2.6)$$

Proof. First, we use the definition of $(L\varphi)(\underline{\eta})$ and expand the expectation with respect to the jump rates to see that

$$\begin{aligned} (L\varphi)\underline{\eta} &= \lim_{\varepsilon \searrow 0} \frac{\mathbb{E}(\varphi(\underline{w}(\varepsilon)) | \underline{w}(0) = \underline{\eta}) - \varphi(\underline{\eta})}{\varepsilon} \\ &= \lim_{\varepsilon \searrow 0} \frac{\sum_{i=a-1}^b (p(\eta_i, \eta_{i+1}) \cdot \varepsilon + o(\varepsilon)) \cdot \varphi(\underline{\eta}^{i,i+1})}{\varepsilon} + \\ &\quad + \lim_{\varepsilon \searrow 0} \frac{\left(1 - \sum_{i=a-1}^b (p(\eta_i, \eta_{i+1}) \cdot \varepsilon + o(\varepsilon))\right) \cdot \varphi(\underline{\eta}) - \varphi(\underline{\eta})}{\varepsilon} \end{aligned}$$

But we know the holding times, T , for some $\lambda := p(w_i, w_{i+1})$, a rate value, are *exponentially distributed* (ie $T \sim \text{Exp}(\lambda)$). Thus, we have the Taylor approximation condition,

$$\mathbb{P}(T < \varepsilon) = 1 - e^{-\lambda\varepsilon} \approx 1 - (1 - \lambda\varepsilon + o(\varepsilon)) = \lambda\varepsilon + o(\varepsilon) \quad (2.7)$$

such that $\lambda := p(w_i, w_{i+1})$. From inserting λ appropriately, we see that

$$(L\varphi)(\underline{\eta}) = \sum_{i=a-1}^b p(\eta_i, \eta_{i+1}) (\varphi(\underline{\eta}^{i,i+1}) - \varphi(\underline{\eta})) \quad (2.8)$$

Now, using the limiting definition of the derivative and the tower rule of conditional expectation, and the above equality,

$$\begin{aligned} \frac{d}{dt} \mathbb{E}(\varphi(\underline{w}(t))) &= \lim_{\varepsilon \searrow 0} \frac{\mathbb{E}(\varphi(\underline{w}(t+\varepsilon)) | \underline{w}(t))}{\varepsilon} \\ &= \lim_{\varepsilon \searrow 0} \mathbb{E} \left(\frac{\mathbb{E}(\varphi(\underline{w}(t+\varepsilon)) - \varphi(\underline{w}(t)))}{\varepsilon} \middle| \underline{w}(t) \right) \\ &= \lim_{\varepsilon \searrow 0} \mathbb{E} \left(\frac{\mathbb{E}(\varphi(\underline{w}(t+\varepsilon)) | \underline{w}(t)) - \varphi(\underline{w}(t))}{\varepsilon} \right) \\ &= \mathbb{E}(L\varphi)(\underline{w}(t)) \end{aligned}$$

□

Corollary 2.8. *A misanthrope process, $\underline{w}$ is on a stationary invariant distribution if and only if the expected value of the infinitesimal generator on a test function is 0.*

Proof. Using the statement in Liggett [7], and Theorem 2.8, setting $\frac{d}{dt} \mathbb{E}(\varphi(\underline{w}(t)))$ to 0 means $0 = \frac{d}{dt} \mathbb{E}(\varphi(\underline{w}(t))) = \mathbb{E}((L\varphi)(\underline{w}(t)))$. $\square$

This means that on some bounded cylinder function, we only need to find expectations of infinitesimal generators that equal zero to find a stationary distribution. This become useful since the expectation of the infinitesimal generator is an intrinsic property of the process, and we know how to condition against it.

Example 2.9. *Totally Asymmetric Simple Exclusion Process (TASEP) Let $w_i \in \Omega$ such that $\Omega := \{0, 1\}$, and p denote right jump rates such that $p(0,1)=1$, and $p(0,0)=p(1,0)=p(1,1)=0$, and q denote left jump rates such that $q(\cdot, \cdot) \equiv 0$. By the construction of p , it is accurate to say that*

$$p(w_i, w_{i+1}) = w_i(1 - w_{i+1}) \quad (2.9)$$

If we denote $\hat{w} := \underline{w}^{i,i+1}$, such that, recalling $\underline{\eta}^{i,i+1}$ from Theorem 2.8, $w_i = 1 \implies \hat{w}_i = 0, w_{i+1} = 0 \implies \hat{w}_{i+1} = 1$

Then TASEP is a kind of misanthrope process, so using Corollary 2.9, for $\underline{w}$ a stationary measure,

$$\begin{aligned} \mathbb{E}(L\varphi)(\underline{w}) &= \mathbb{E} \sum_{i=a-1}^b w_i(1 - w_{i+1})(\varphi(\underline{w}^{i,i+1}) - \varphi(\underline{w})) \\ &= \mathbb{E} \sum_{i=a-1}^b (1 - \hat{w}_{i+1})\hat{w}_{i+1}(\varphi(\hat{w}) - w_i(1 - w_{i+1})\varphi(\underline{w})) \end{aligned}$$

Now, we change variables of the expectation, to have it with respect to w_i from $\hat{w}_i$ for all i . In this case, for the measure μ , the Radon-Nikodym derivative, $\frac{\mu(\hat{w}_{i+1})\mu(\hat{w}_{i+1}-1)}{\mu(\hat{w}_i)\mu(\hat{w}_{i+1})}$, is 1, since we have not had any scaling in the change, only translation. Thus

$$\begin{aligned} \mathbb{E}(L\varphi)(\underline{w}) &= \mathbb{E} \sum_{i=a-1}^b (1 - w_i)w_{i+1} - w_i(1 - w_{i+1}) \cdot \varphi(\underline{w}) \\ &= \mathbb{E} \left(\sum_{i=a-1}^b (w_{i+1} - w_i) \varphi(\underline{w}) \right) \\ &= \mathbb{E}((w_{b+1} - w_{a-1})\varphi(\underline{w})) \\ &= \mathbb{E}(w_{b+1} - w_{a-1})\mathbb{E}(\varphi(\underline{w})) \end{aligned}$$

Since w_{b+1}, w_{a-1} are independent of $\varphi(\underline{w})$.

This all means that knowing the expectation of the infinitesimal generator gives us knowledge of the expected value of a the bounded cylinder function and the difference of values just outside the bounded set, and so we can extend our

knowledge to all values of the interacting particle system inductively, only knowing a finite amount. Hence, all we need to know, to produce a stationary distribution on the TASEP, is w_{b+1} , w_{a-1} , and $\mathbb{E}(\varphi(\underline{w}))$.

2.3 The translation invariant product distribution

Referring back to $f: S \rightarrow \mathbb{R}$ as in the symmetric property, we have this statement.

Remark 2.10. *The law of attractivity means that f is non-decreasing on S .*

Theorem 2.11. *In the symmetric property, for the function f , for any $x, y, z \in \Omega$, $\frac{f(z)}{f(x)}$ does not depend on y .*

Proof. We have that, by the symmetric property, $p(y, z - 1) = s_p(y, z)f(y)$. This implies

$$\begin{aligned} \frac{p(y, z - 1)}{f(y)} &= s_p(y, z) = s_p(z, y) = \frac{p(z, y - 1)}{f(z)} \\ &\implies \frac{f(z)}{f(y)} = \frac{p(z, y - 1)}{p(y, z - 1)} \\ &\implies \log(f(z)) - \log(f(y)) = \log(p(z, y - 1)) - \log(p(y, z - 1)) \\ &\quad \text{and} \\ &\log(f(y)) - \log(f(x)) = \log(p(y, x - 1)) - \log(p(x, y - 1)) \end{aligned} \tag{2.10}$$

So, $\log(f(z)) - \log(f(x)) = b + a$ and we are done. $\square$

Now, we define a product measure that is a translation invariant stationary product distribution, as in Balazs and Seppäläinen [1], which takes ideas from Coccozza-Thivent [3].

Choose a function $Z : S \rightarrow \mathbb{R}$ such that

$$Z(\theta) = \sum_{y=w^{\min}}^{w^{\max}} \frac{e^{\theta y}}{f(y)!} \tag{2.11}$$

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3 Result

3.1 ASEP Example

First, let us introduce the framework for the result in the ASEP case, with the generator $L = \mathcal{L} + \mathcal{R}$, and constant left and right rates $p, q \in \mathbb{R}$ such that $p + q = 1$, choosing $q \leq p$. The stationary measure μ is such that $\mu(\mathcal{L} + \mathcal{R}) = 0$. Since $\mathcal{L}$ and $\mathcal{R}$ are just the same process scaled to q and p , it follows that $\mu\mathcal{L} = 0$ and $\mu\mathcal{R} = 0$.

Taking $S = \frac{\mathcal{L} + \mathcal{L}^*}{2}$, S is a symmetric process. See also, since in ASEP,

$$L^* = \mathcal{L}^* + \mathcal{R}^* = \frac{q}{p}\mathcal{R} + \frac{p}{q}\mathcal{L} \quad (3.1)$$

we have

$$S = \frac{\left(\frac{p}{q} + 1\right)\mathcal{L} + \left(\frac{q}{p} + 1\right)\mathcal{R}}{2} = \frac{\frac{1}{q}\mathcal{L} + \frac{1}{p}\mathcal{R}}{2} \quad (3.2)$$

To find the maximally asymmetric process that has the same stationary distribution, we examine

$$A = L - S = \frac{\mathcal{L} + \mathcal{L}^*}{2} \quad (3.3)$$

which describes the furthest p and q can be from each other and still have the same stationary distribution. This is not a generator, however, since $q \leq p$. Thus we create, for $\alpha \in \mathbb{R}$,

$$G = L = \alpha S = \mathcal{L} - \frac{\alpha}{2q}\mathcal{L} + \mathcal{R} - \frac{\alpha}{2p}\mathcal{R} \quad (3.4)$$

such that α is large as possible with G still being a generator. Since $q \leq p$, we take $\alpha = 2q$, so

$$G = \left(1 - \frac{2q}{2p}\right)\mathcal{R} = \frac{p - q}{p}\mathcal{R} \quad (3.5)$$

where G has right jump rate coefficient $p - q$, and is TASEP. This is the **maximally asymmetric stationary invariant generator** for the ASEP process.

3.2 3 State Misanthrope Process

As we can see from the graphs produced in Maple, the three distinct functions bounding α are all monotone for $p \in [0.5, 1]$ and $k \geq 1$.

$$\begin{aligned} y_3 = y_2 = y_1 & \quad \text{for} \quad k = 1, p = 0.5 \\ y_3 = y_2 < y_1 & \quad \text{for} \quad k = 1, p \in (0.5, 1] \\ y_3 < y_2 < y_1 & \quad \text{for} \quad k > 1, p \in (0.5, 1] \end{aligned}$$

Furthermore, $\lim_{k \rightarrow \infty} y_1 = 2$, and $\lim_{k \rightarrow \infty} y_3 = 0$, so it is safe to say

$$\alpha = \frac{4(1-p)}{2+p(k-1)} = \frac{4(1-p)}{2+p\left(\frac{a}{c}-1\right)} = \frac{4c-4pc}{2c+pa-pc} \quad (3.6)$$

With this in mind, we have that, for the **right** flow, the infinitesimal generator $L - \alpha S =$ has rates

$$\begin{aligned} \rho(0,0) &= p(0,0) - \alpha P(0,0) = pc - \frac{4c-4pc}{2c+pa-pc} \frac{c}{2} = c \left(p + 2c \frac{p-1}{c(2-p)+ap} \right) \\ \rho(0,-1) &= p(0,-1) - \alpha P(0,-1) = pc - \frac{4c-4pc}{2c+pa-pc} \frac{c}{2} = c \left(p + 2c \frac{p-1}{c(2-p)+ap} \right) \\ \rho(1,0) &= p(1,0) - \alpha P(1,0) = \frac{pc}{2} + \frac{pa}{2} - \frac{4c-4pc}{2c+pa-pc} \left(\frac{c}{4} + \frac{pa+(1-p)c}{4} \right) = \\ &= \frac{p(a+3c)}{2} - c \\ \rho(-1,-1) &= p(1,-1) - \alpha P(1,-1) = pa - \frac{4c-4pc}{2c+pa-pc} \frac{a}{2} = 2ac \left(p - \frac{1-p}{2c+pa-pc} \right) \end{aligned}$$

and the **left** flow

$$\begin{aligned} \lambda(0,0) &= q(0,0) - \alpha Q(0,0) = (1-p)c - \frac{c}{2} \frac{4c-4pc}{2c+pa-pc} = c(1-p)p \frac{a-c}{2c+pa-pc} \\ \lambda(-1,0) &= q(-1,0) - \alpha Q(-1,0) = (1-p)c - \frac{c}{2} \frac{4c-4pc}{2c+pa-pc} = c(1-p)p \frac{a-c}{2c+pa-pc} \\ \lambda(0,1) &= q(0,1) - \alpha Q(0,1) = (1-p)c - \left(\frac{c}{4} - \frac{(1-p)c+pa}{4} \right) \frac{4c-4pc}{2c+pa-pc} = 0 \\ \lambda(-1,1) &= q(-1,1) - \alpha Q(-1,1) = (1-p)a - \frac{a}{2} \frac{4c-4pc}{2c+pa-pc} = a(1-p)p \frac{a-c}{2c+pa-pc} \end{aligned}$$

And this, by virtue of being a linear combination of L and S , has the same stationary invariant distribution, μ , as L (Computations to prove this directly are in the appendix). Thus we have an example of a 3 state misanthrope IPS who's left and right jump rates, independently, are not stationary invariantly distributed, but who's union is. Furthermore, we have an IPS of this type that is *maximally* asymmetric.

3.3 Proving non-trivial stationary invariance

We see if the 4 conditions for a stationary invariant distribution hold

3.3.1 Absorption at endpoints

$$\rho(w^{\min}, \cdot) \equiv \rho(\cdot, w^{\max}) \equiv \lambda(w^{\max}, \cdot) \equiv \lambda(\cdot, w^{\min}) \equiv 0 \quad (3.7)$$

Holds by $L - \alpha S$ being constructed of symmetric L and S

- 3.3.2 Law of attraction
- 3.3.3 Cyclic property
- 3.3.4 Symmetric property

4 Appendix

4.1 A

$$(L\varphi)\underline{\eta} = \frac{\lim_{\varepsilon \searrow 0} \mathbb{E}(\varphi(w(\varepsilon)) | \underline{w}(0) = \underline{\eta}) - \varphi(\underline{\eta})}{\varepsilon} = \lim_{\varepsilon \searrow 0} \sum_{i=a-1}^b \dots \quad (4.1)$$



Figure 2: The Universe

5 Conclusion

“I always thought something was fundamentally wrong with the universe”
[1]

References

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